

A METHOD FOR ASSESSING EFFICIENCY AND BIAS OF ESTIMATION STRATEGIES IN SURVEY SAMPLING

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SUMMARY

The now traditional setup for survey sampling with labelled units drawn by arbitrary, unequal probability sampling designs is interpreted in this paper in a continuous, infinite population framework. This is used to study efficiencies of various sampling strategies under a super-population model with an auxiliary variable:

$$E_{\xi} \{Y(x)|x\} = \Theta x^b, \quad V_{\xi} \{Y(x)|x\} = \sigma^2 x^g.$$

We study the effects of population shape, especially skewness, on the efficiency of various procedures. The extent of the bias resulting from model misspecification is also investigated.

1. The continuous survey sampling model

Consider first a finite population of size N where y is the variable of interest, whose mean we wish to estimate, and x is an auxiliary variable. An unknown number, y_i , and a known number x_i , are associated with the unit identified by the label i ($i=1, \dots, N$). If one unit is drawn from this population, the observation can be recorded as the pair (y_i, i) . If the numbers x_i are distinct, exactly the same information is carried by the pair (y_i, x_i) . A unit can thus be identified by its label value or, equivalently, by its auxiliary variable value. As is often pointed out the label itself is an auxiliary variable. This setup that emphasizes the role of the labels is due to Godambe (1955, 1965); a recent discussion is found in C.R. Rao (1977).

The continuous variable formulations, Cassel and Särndal (1972, 1974) and the extensions in this paper are an attempt to steep Godambe's survey sampling setup in continuous terms. Such an interpretation makes it easier to grasp some of the complexities of modern survey sampling theory; exact efficiencies of various sampling strategies can often be computed, and the effect of population skewness and other characteristics can be studied.

Consider now a population of infinitely many pairs $(y(x), x)$, where $x \geq 0$, with unknown joint distribution, but with known marginal distribution of x . Let $F(x)$ be the

probability, consider known for any $x \geq 0$, that the x -component of a randomly chosen pair $(y(x), x)$ does not exceed x , and let $F(x) = \int_0^x f(x) dx$. The moments of x ,

$$\mu_{Xb} = E_f(X^b) = \int_0^\infty x^b dF(x)$$

are therefore known quantities. (This expectation defines the operator E_f . Two additional operators E_p and E_ξ will be needed later.) For simplicity, write μ_X for μ_{X1} .

In the continuous approach, the label of a population unit is a continuous index λ , where for convenience $\lambda \in [0, 1]$, as in Cassel and Särndal (1972, 1974). The linking of a label to its x -value (the "labelling") can be expressed by any suitable function. We shall more specifically impose an ordering: We define λ as the λ -quantile of the x -distribution, so that $\lambda = F(x)$. Solving for x , let $x = h(\lambda)$, where $h(\lambda) = F^{-1}(\lambda)$ is the inverse function of $F(x)$.

Having drawn and observed one unit, we can record the data as $(y_1(\lambda), \lambda)$ or, equivalently, as $(y(x), x)$. In the latter case, the unit drawn is identified by its x -value. The two functions, $y_1(\cdot)$ and $y(\cdot)$, are related by $y_1(\lambda) = y[h(\lambda)]$. The problem to be studied is that of estimation of the population mean of Y ,

$$(1.1) \quad m_Y = E_f(Y) = \int_0^\infty y(x) dF(x) = \int_0^1 y\{h(\lambda)\} d\lambda$$

Recall that Godambe's (1965) finite setup defines a sampling design as a real function p on S , the set of subsets s of $\{1, \dots, N\}$, such that $0 \leq p(s) \leq 1$ for all $s \in S$ and $\sum_{s \in S} p(s) = 1$.

In the continuous setup and in the presence of an auxiliary variable x , members of a sample of fixed size n are identified either by the vector of labels drawn, $\lambda = (\lambda_1, \dots, \lambda_n)$, where $0 \leq \lambda_i \leq 1$ ($i=1, \dots, n$), or by the corresponding vector $x = (x_1, \dots, x_n)$, where $\lambda_i = F(x_i)$ ($i=1, \dots, n$). The y -value is observed for each of the labels drawn, and we can record the data as $(y(x_i), x_i)$ ($i=1, \dots, n$).

Let S be the σ -algebra of subsets of $\mathcal{R}_+^n = \{x | x_i \geq 0 \text{ (} i=1, \dots, n \text{)}\}$. A sampling design is defined as a real, non-negative function Q on S such that $\int_{\mathcal{R}_+^n} dQ = 1$, where $Q(x)$ is the probability of drawing a sample such that the auxiliary variable value in the i :th draw does not exceed x_i ($i=1, \dots, n$). Let $q(x) = dQ(x)/dx$ be the joint density function, and assume that $q(x) = p(x)f(x)$, where $f(x) = \prod_{i=1}^n f(x_i)$.

$$\int_{\mathcal{R}_+^n} q(x) dx = \int_{\mathcal{R}_+^n} p(x) f(x) dx = \int_{A^n} p\{h(\lambda_1), \dots, h(\lambda_n)\} d\lambda_1 \dots d\lambda_n$$

where $p\{h(\lambda_1), \dots, h(\lambda_n)\}$ is a density in $\lambda_1, \dots, \lambda_n$ on $A^n = \{\lambda: 0 \leq \lambda_i \leq 1 (i=1, \dots, n)\}$.

We shall refer to the function $p(x)$ hereby introduced as the *design function*. Since a unit can be identified by its x -value, the sampler can choose his function $p(x)$ suitably.

Now, the case where $p(x) = 1$ for all $x \in R_+^n$, which corresponds to the mathematical statistician's standard assumption of "independent, identically distributed" observations $x_i (i=1, \dots, n)$, represents unrestricted simple random sampling (srs) from the population and then $q(x) \equiv f(x)$. Otherwise, $q(x) \neq f(x)$. In general, $q(x)$ is the density generated by the randomization device $p(x)$. Godambe and Thompson (1971) used the term "frequency distribution generated by randomization" with a similar connotation.

Let $q_i(x) = p_i(x)f(x)$ be the marginal density of the i :th component of x . The expected x -value in draw i is

$$E_{p_i}(X) = \int_0^\infty x p_i(x) f(x) dx$$

(The expectation operator E_{p_i} is with respect to the i^{th} marginal density in x , $p_i(x)f(x) = q_i(x)$. E_p will denote expectation with respect to the joint design density, $p(x)f(x) = q(x)$.) The density of the label $\lambda = F(x)$ obtainable in the i :th draw is $p_i[h(\lambda)]$, so that the expected label in draw i is

$$E_{p_i}(\lambda) = \int_0^1 F(x) p_i(x) f(x) dx = \int_0^1 \lambda p_i\{h(\lambda)\} d\lambda.$$

For example, if $p_i(x) \equiv 1$, then $E_{p_i}(\lambda) = 0.5$, which simply means that the population median is what we expect to draw under srs.

An example of a design function producing draws with correlated labels is

$$(1.2) \quad p(x) = \frac{\Gamma(s)}{\prod_{i=1}^{n+1} \Gamma(sd_i)} \prod_{i=1}^{n+1} \{F(x_i) - F(x_{i-1})\}^{sd_i - 1}$$

for $0 = x_0 < x_1 < \dots < x_n < x_{n+1} = \infty$, where $s > 0$, $d_i = \alpha_i - \alpha_{i-1} > 0 (i=1, \dots, n+1)$, and $\alpha_0 = 0$, $\alpha_{n+1} = 1$, so that $\sum_{i=1}^{n+1} d_i = 1$. Here, s and $0 < \alpha_1 < \alpha_2 < \dots < \alpha_n < 1$ are specified values. Under the randomization device (1.2), the expected label in draw i is $E_{p_i}(\lambda) = \alpha_i$, regardless of the population shape $F(x)$. The variance of the label in draw i is $\alpha_i(1 - \alpha_i)/(s+1)$, and the covariance between the labels in draws i and j is $-\alpha_i(\alpha_j - \alpha_i)/(s+1)$, for $\alpha_i < \alpha_j$. At the design stage, the statistician may fix the constants s and $\alpha_1, \dots, \alpha_n$, suitably spaced over the $(0,1)$ interval. For example, a concentration of α -values near unity would favour inclusion of units with large x -values.

In this paper we give an example of the use of this framework. We wish to study a broad family of population shapes and of design functions, while maintaining mathematical simplicity.

This goal is met if we assume that individual draws are independent, with possibly unequal marginal design functions, so that $p(\mathbf{x}) = \prod_{i=1}^n p_i(x_i)$, where $p_i(x_i) \propto x_i^{m_i-1}$, and that sampling is from a gamma(f) population with density $f(x) = x^{f-1} e^{-x} / \Gamma(f)$ for $x > 0$. Then a convenient amalgamation of $p_i(x_i)$ and $f(x)$ obtains: Letting

$$p_i(x) = \Gamma(f) x^{m_i} / \Gamma(f+m_i),$$

the generated density $q_i(x)$ is the gamma($f+m_i$) density, provided $m_i > -f$. Hence, the population mean of X is $\mu_X = E_f(X) = f$, while the expected x -value in the i^{th} draw is $E_{p_i}(X) = f+m_i$. Here the expected label in draw i depends on the population shape; for example, if $f = 1$, $E_{p_i}(\lambda) = 1 - 2^{-m_i-1}$; if $f = 2$, $E_{p_i}(\lambda) = 1 - (m_i+4)2^{-m_i-3}$.

Our example involves the practically important superpopulation model ξ :

$$(1.3) \quad \begin{aligned} Y(x) &= \Theta x^b + Z(x); E_{\xi} \{Z(x)|x\} = 0; \\ E_{\xi} \{[Z(x)]^2|x\} &= \sigma^2(x); E_{\xi} \{Z(x_1)Z(x_2)|x_1, x_2\} = 0 \quad (x_1 \neq x_2) \end{aligned}$$

Here, $\Theta > 0$ is unknown, and $b \geq 0$ may or may not be known. This model together with the heteroscedasticity assumption $\sigma^2(x) = \sigma^2 x^g$ will be referred to as the (b, g) -model. Any particular infinite population value, $y(x) = \Theta x^b + z(x)$, is to be interpreted as Θx^b plus a realization of the random variable $Z(x)$. Defining $\mu_Y = E_f E_{\xi}[Y(x)]$, we have $\mu_Y = \Theta \mu_{X^b}$ and $E_{\xi}(m_Y) = \mu_Y$, where m_Y is given by (1.1). In Section 3 we let $b = 1$; in Section 4 we study the bias that results from acting as if $b = 1$ although in reality $b \neq 1$. As for g , assume $0 \leq g \leq 2$, which covers most situations of practical interest, as pointed out by Rao and Bayless (1969). The correlation between $y(x)$ and x is

$$\rho = E_f E_{\xi} (Y(x) - \mu_Y)(x - \mu_X) / \{E_f E_{\xi} (Y(x) - \mu_Y)^2 E_f (x - \mu_X)^2\}^{1/2}$$

For a gamma(f) population, letting $k = \sigma/\Theta$, we have

$$(1.4) \quad \rho^2 = 1 / \{1 + k^2 \Gamma(f+g) / \Gamma(f+1)\}$$

For the continuous setup in conjunction with the model (1.3), the population can thus be thought of in terms of a bivariate scatter with an infinite number of (x, y) -points, where the marginal density of x is $f(x)$ and where, for each fixed x -value, the y -value has been generated by (1.3).

2. Estimators and Strategies

Let T be an estimator of m_Y . Using current terminology in sampling literature we shall say that: (a) T is p -unbiased if, for all m_Y , $E_p(T) = m_Y$; (b) T is ξ -unbiased if, for all x , $E_\xi(T - m_Y) = 0$; (c) T is $p\xi$ -unbiased if, for all μ_Y , $E_p E_\xi(T) = \mu_Y$. Either p -unbiasedness or ξ -unbiasedness implies $p\xi$ -unbiasedness. An estimation strategy will be denoted (P, T) ; it is the combination of a design $p = p(x)$ and an estimator T . We look only at $p\xi$ -unbiased strategies; in addition we want the variance to be small under a range of conditions. Many interesting estimators have the form

$$T = \mu_{Xb_0} \frac{\sum x_i^a y(x_i)}{\sum x_i^{a+b_0}}$$

For any a , this estimator is ξ -unbiased under a (b, g) -model with $b = b_0$. (All sums are over i from 1 to n .) Of this form is

$$(2.1) \quad T = \mu_{Xb_0} \frac{\sum x_i^{b_0 - g_0} y(x_i)}{\sum x_i^{2b_0 - g_0}}$$

which, by generalized least squares theory, is best linear ξ -unbiased under a (b, g) -model with $b = b_0, g = g_0$. But this optimality is "weak": it is retained only if (b_0, g_0) is truly the value of (b, g) .

In practice, the statistician is not likely to know (b, g) but assumes $(b, g) = (b_0, g_0)$, say. Hence, he may wish to consider (2.1) as his estimator. If the assumption $b = b_0$ is correct, but $g = g_0$ is false, then (2.1) is ξ -unbiased but not best ξ -unbiased. The loss of the "bestness" does not always create a serious loss of efficiency. Worse is if $b = b_0$ is a wrong assumption: Then the estimator (2.1) is not even $p\xi$ -unbiased.

By (2.1), the usual Ratio and Mean-of-the-Ratios estimates,

$$(2.2) \quad T_R = \mu_X \frac{\sum y(x_i)}{\sum x_i}$$

$$(2.3) \quad T_{MR} = \frac{\mu_X}{n} \sum \frac{y(x_i)}{x_i}$$

are best ξ -unbiased under the (b, g) -model with $b = 1, g = 1$ and $b = 1, g = 2$, respectively. Under certain conditions, the variances of T_R and T_{MR} become progressively smaller as the design function shifts in the direction of assigning ever larger inclusion probability to large x -values; see Remark 1 below. This corresponds to the known fact that, in the finite setup, purposive selection of the n units with the largest x -values is sometimes optimal, Brewer (1963), Royall (1970). Unfortunately, the more one emphasises inclusion of the large- x units, the larger will be the $p\xi$ -bias if $b \neq 1$, see Section 4.

In continuous form, the p -unbiased and $p\xi$ -unbiased estimator of Horvitz and Thompson (1952) becomes

$$(2.4) \quad T_{HT} = \frac{1}{n} \sum \frac{y(x_i)}{p(x_i)}$$

(The distinction between with and without replacement is not an issue in the continuous approach: There is always zero probability of selecting the same x -value more than once.) If $p_i(x) = 1$ ($i = 1, \dots, n$), (2.4) reduces to the sample mean per unit,

$$\bar{Y}_s = \sum y(x_i)/n$$

Optimality properties have been established for T_{HT} under certain models implying $g = 2$: C.R. Rao (1971), Godambe and Thompson (1973), J.N.K. Rao (1973), Cassel, Särndal and Wretman (1977). If $g = 2$, then T_{HT} with $p_i(x) \propto x$ ($i = 1, \dots, n$) is certainly a very strong estimator, as shown by the efficiency calculations in Section 3.

We also consider stratified sampling involving L strata, where a given unit belong in stratum h if its x -value is in the interval (x_{h-1}, x_h) ($h = 1, \dots, L$) where

$$0 = x_0 < x_1 < \dots < x_L = \infty$$

The traditional p -unbiased and $p\xi$ -unbiased estimator takes the form

$$T_{ST} = \sum_{h=1}^L W_h \bar{y}_h$$

where, letting n_h be the number of units sampled from stratum h ,

$$W_h = F(x_h) - F(x_{h-1}); \bar{y}_h = \sum_{i=1}^{n_h} y(x_i)/n_h$$

We shall evaluate the performance of certain strategies involving the estimators introduced above. Here, $(x^{\frac{m}{1}}, T)$ will denote the strategy consisting of the estimator T

used together with independent draws according to $p_i(x) \propto x^{m_i}$ ($i=1, \dots, n$). If draws are independent with identical design functions $p_i(x) \propto x^{m_i}$ ($i=1, \dots, n$), we write the strategy as (x^m, T) . Also, let us write (srs, T) and (pps, T) when, respectively, $p_i(x) = 1$ ($i=1, \dots, n$), and $p_i(x) \propto x$ ($i=1, \dots, n$); the latter represents "probability proportional to size" sampling. Let srs denote the design of srs within each of L strata.

The quality of an estimation strategy will be evaluated by its $p\xi$ -Mean squared error,

$$MSE(p, T) = E_p E_\xi \{(T - \mu_Y)^2\} = \text{Var}(p, T) + \{\text{bias}(p, T)\}^2$$

where

$$\text{Var}(p, T) = E_p E_\xi \{T - E_p E_\xi(T)\}^2; \text{bias}(p, T) = E_p E_\xi(T) - \mu_Y$$

This measure refers to a specified model and a specified sampling design. The MSE is usually a complicated expression but it can sometimes be evaluated in closed form using techniques similar to those in Rao and Webster (1966). Some results are given in the next section.

3. Strategy Evaluation when $b = 1$

In this section we assume that the population conforms to a (b, g) -model with $b = 1$. All strategies considered are $p\xi$ -unbiased, but some are p -biased. Sampling is from the gamma(f) population. Draws are independent with design functions $p_i(x) \propto x^{m_i}$. Hence, $q_i(x)$ is the gamma(r_i) density, where $r_i = f + m_i$. Let $\bar{m} = \sum m_i/n$.

For the ratio estimator (2.2), we get

$$(3.1) \quad \text{Var}(x^{m_i}, T_R) = \sigma_X^2 \frac{\sum \{\Gamma(r_i + g)/\Gamma(r_i)\}}{(g-1 + \sum r_i)(g-2 + \sum r_i)}$$

provided $g - 2 + \sum r_i > 0$. When $g = 0, 1$ and 2 it is easily shown that, for a fixed $\bar{m}$, (3.1) attains a minimum if $m_i = \bar{m}$ ($i = 1, \dots, n$), so no variance reduction is obtained by "diversifying" the design through use of different m_i . In the special case of srs (all $m_i = 0$), (3.1) agrees with a result of P.S.R.S. Rao (1968). Our approach may also be used to extend to arbitrary $p_i(x) \propto x^{m_i}$ the results by J.N.K. Rao (1967), P.S.R.S. Rao and J.N.K. Rao (1971) for various variations of the ratio estimator. Their work assumed srs and an infinite gamma population.

For the Mean-of-the-Ratios estimator (2.3), the variance is given by

$$(3.2) \quad \text{Var}(x^{m_i}, T_{MR}) = \frac{\sigma^2 \mu_X^2}{n^2} \sum \frac{\Gamma(r_i + g - 2)}{\Gamma(r_i)}$$

When $g = 2$, then (3.2) reduces to $\sigma^2 \mu_X^2 / n$ regardless of population shape and design function. When $g = 0$ and $g = 1$, it is again easy to show that nothing is gained by "diversifying" the design: For a fixed value $\bar{m}$, the minimum variance obtains when $m_i = \bar{m}$ ($i = 1, \dots, n$).

If $m_i = m$ ($i = 1, \dots, n$), then (3.1) and (3.2) reduce to

$$(3.3) \quad \text{Var}(x^m, T_R) = \frac{\sigma^2 \mu_X^2 C_n}{n} \frac{\Gamma(r+g)}{r\Gamma(r+1)}$$

$$(3.4) \quad \text{Var}(x^m, T_{MR}) = \frac{\sigma^2 \mu_X^2}{n} \frac{\Gamma(r+g-2)}{\Gamma(r)}$$

where $r = f + m$, and $C_n = 1 / \{1 + (g-1)/nr\} \{1 + (g-2)/nr\}$. Here, C_n tends to 1 as n increases and the approximation $C_n \simeq 1$ is good even when n is moderately large.

Remark 1:

The effect of giving high inclusion probability to units with large x -value: If $g > 0$, these units are the ones with high model variance. A large m -value produces large inclusion probabilities for these units. For fixed n , the effect of letting m increase is: If $g < 2$, both (3.3) and (3.4) decrease towards zero. If $g = 2$, $\text{Var}(x^m, T_R)$ decreases towards $\sigma^2 \mu_X^2 / n = \text{Var}(x^m, T_{MR})$. Thus, if $g < 2$, an increase in m has a variance decreasing effect akin to that of increasing the sample size.

Remark 2:

The effect of population skewness: The variance-reducing effect due to an increase in m is, for both T_R and T_{MR} , more pronounced for skewed populations. Compare the cases $m = 6$ and $m = 1$, when $g = 1$: If $f = 1$ (exponential shape), then

$$\text{Var}(x^6, T_{MR}) / \text{Var}(x^1, T_{MR}) = 1/6,$$

whereas if $f = 5$ (less skewed population), the same ratio of variances is $1/2$ only. If population skewness is considerable, (x^m, T_{MR}) may have a very large or infinite variance; it is infinite if $r + g - 2 \leq 0$. The strategy (x^m, T_R) is less sensitive to skewness, its variance being finite as soon as $n > (2-g)/r$.

From (3.3) and (3.4), we derive the two efficiencies;

$$(3.5) \quad \frac{\text{Var}(x^m, T_R)}{\text{Var}(x^m, T_{MR})} \simeq \{1+(g-1)/r\} \{1+(g-2)/r\}$$

$$(3.6) \quad \frac{\text{Var}(x^m, T_R)}{\text{Var}(x^{m+1}, T_{MR})} \simeq 1 + (g-1)/r$$

where the approximation comes about by setting $C_n \simeq 1$ in (3.3).

It is known (Royall, 1970) that if T_R and T_{MR} are used with the same design, then (p, T_R) is a better strategy than (p, T_{MR}) if $g \leq 1$, and the reverse is true if $g \geq 2$. For any gamma(r) population, our result (3.5) shows not only which strategy is preferable if $1 < g < 2$, but also gives the value of the efficiency of one strategy relative to the other, for any g and any design $p_i(x) \propto x^m$ ($i = 1, \dots, n$). Foreman and Brewer (1971) showed that (pps, T_{MR}) is better than (srs, T_R) if $g > 1$. Under almost identical conditions, (3.6) confirms their result. In addition, we obtain a value of the efficiency of one strategy relative to the other.

Remark 3:

From the comparison (3.5), in which both estimators are used with the same design, we see that (x^m, T_R) is more efficient than (x^m, T_{MR}) as long as $g \leq g_0$, otherwise (x^m, T_{MR}) is better. Now, g_0 is somewhat above 1.5, varying slightly with $r = f + m$: If $r = 1$, then $g_0 = 1.62$; if $r = 5$, $g_0 = 1.52$.

Remark 4:

The efficiency (3.6) is relevant, for example, in the comparison of (srs, T_R) vs. (pps, T_{MR}) . If $g = 1$, (3.6) shows that the two strategies are about equally efficient. If $g < 1$, (x^m, T_R) is more efficient; if $g > 1$, (x^m, T_{MR}) is the more efficient strategy. Again, population skewness is a decisive factor in the comparison: If g is about 0 or about 2, the efficiency gain realized by the better of the two strategies may run from substantial (for markedly skew populations, say, $f = 1$) to small or insignificant (for almost symmetrical populations, $f = 10$, say).

The Horvitz-Thompson estimator, always p -unbiased and thus $p\bar{\xi}$ -unbiased, has the variance

$$(3.7) \quad \text{Var}(x^m, T_{HT}) = \frac{\Theta^2 \mu_X^2}{n} \left\{ \frac{\Gamma(f+m) \{ \Gamma(f-m+2) + k^2 \Gamma(f-m+g) \}}{\{ \Gamma(f+1) \}^2} - 1 \right\}$$

where $k = \sigma/\Theta$. Given f , this is a U-shaped function of m . Outside a narrow range of m -

values, the variance is large; it remains finite only if m satisfies $-f < m < f + \min(2, g)$. The variance is particularly sensitive to changes in m when the population is skewed; compare Basu's (1971) "Circus example". If the bottom of the U occurs near $m = 0.5$, then $(\sqrt{x}, T_{HT})$ can be considerably more efficient than either the $m = 0$ case, $(srs, T_{HT}) \equiv (srs, \bar{Y})$, or the $m = 1$ case $(pps, T_{HT}) \equiv (pps, T_{MR})$. In Table 1, this occurs, for example, if $g = f = 1$, $k = 5$. The fact that sampling with probability proportional to square root of size is sometimes beneficial was recognized by Hansen and Hurwitz (1949); see also T.J. Rao (1971).

Setting $m = 0$ in (3.7), we note, using (1.3) and (3.3), that the sample mean per unit has the efficiency

$$\frac{\text{Var}(srs, T_R)}{\text{Var}(srs, \bar{Y}_s)} = (1 - \rho^2) C_n \simeq 1 - \rho^2$$

For a proportionally allocated ($n_h \propto W_h$) stratified sample, using many strata with optimally located stratum boundaries, we get

$$(3.8) \quad \text{Var}(srst, T_{ST}) = \frac{\sigma_X^2 \mu_X^2}{n} \frac{\Gamma(r+g)}{r\Gamma(r+1)}$$

Thus, (3.8) represents the minimum variance obtainable by proportional stratified sampling, but in practice even a moderate number of strata with optimum or near-optimum stratum boundaries brings us to the minimum. From (3.3) and (3.8), we derive:

$$\frac{\text{var}(srs, T_R)}{\text{var}(srst, T_{ST})} = C_n \simeq 1$$

Hence, under these conditions $(srst, T_{ST})$ and (srs, T_R) are roughly equally efficient. This agrees with an informal observation by Cochran (1977, p. 169) who, in reference to optimally allocated stratified sampling, finds that "stratification by size of farm accomplishes the same general purpose as a ratio estimate in which the denominator is farm size."

If $b \neq 1$ in the (b, g) -model then T_R and T_{MR} are $p\xi$ -biased unless used with a design for which the strategy is p -unbiased. This is the case for (pps, T_{MR}) and, approximately, for (srs, T_R) . These "conservative" usages of T_{MR} and T_R are compared in Table 1 with $(srs, \bar{Y}_s)$, $(\sqrt{x}, T_{HT})$ and $(srst, T_{ST})$, which are p - and $p\xi$ -unbiased for any (b, g) -model. Note that $(pps, T_{HT}) \equiv (pps, T_{MR})$ and $(srs, T_{HT}) \equiv (srs, \bar{Y}_s)$.

Remark 1 has shown that this conservative use of T_R and T_{MR} is costly in terms of loss of efficiency. A very small variance could be obtained if the parameter m were cho-

sen to be large. However, a practitioner who argues that T_R should be used together with srs would be justified on the grounds that the bias occurring when b is not equal to one is a more serious consideration than the gain of efficiency realized by, say, (x^5, T_R) when b equals one.

The large sample efficiencies given in Table 1 are relative to (srs, T_R) , that is, the table value is $\text{Var}(srs, T_R)/V$, where V denotes the variance of a competing strategy. The selected values of $k = \sigma/\Theta$ (1/5, 1 and 5) indicate small, moderate, and considerable scatter around the regression line $y = \Theta x$. When $g = 1$, the correlation (1.4) between x and y is 0,98 for $k = 1/5$, but only 0,20 for $k = 5$. In the latter case, the correlation is so weak that $(srs, \bar{Y}_s)$ is almost as efficient as (srs, T_R) .

TABLE 1. Large sample efficiencies of various sampling strategies relative to (srs, T_R) ; $k = \sigma/\Theta$

Strategy		g=0		g=1		g=2	
		f=1	f=5	f=1	f=5	f=1	f=5
$(srs, \bar{Y}_s)$	k = 0,2	0,04	0,01	0,04	0,04	0,07	0,19
	k = 1	0,50	0,17	0,50	0,50	0,67	0,86
	k = 5	0,96	0,83	0,96	0,96	0,98	0,99
$(\sqrt{x}, T_{HT})$	k = 0,2	0,08	0,03	0,19	0,15	0,36	0,54
	k = 1	0,57	0,45	1,04	0,84	1,47	1,10
	k = 5	0,63	0,91	1,26	1,04	1,69	1,14
(pps, T_{HT})	any k	0	0,80	1,00	1,00	2,00	1,20
$(srst, T_{ST})$	any k	1,00	1,00	1,00	1,00	1,00	1,00

Remark 5:

If $g = 0$, (srs, T_R) or $(srst, T_{ST})$ are preferable strategies, and (pps, T_{HT}) is to be avoided, especially if the skewness is considerable. If $g = 1$, (pps, T_{HT}) , (srs, T_R) and $(srst, T_{ST})$ are equally efficient strategies. All three will, however, sometimes be outperformed by $(\sqrt{x}, T_{HT})$. If $g = 2$, (pps, T_{HT}) is clearly better than its competitors.

4. The extent of the bias when the model is false

A model-dependent estimator is usually free of p ξ -bias only under restrictive con-

ditions: Either some specific model has to hold true, or the sampling design has to be of a specific kind. For example, the sample mean per unit $\bar{Y}$ is $p\xi$ -unbiased under any design if a (b, g) -model with $b = 0$ is true, but only under the srs design if $b \neq 0$. Similarly, T_{MR} is $p\xi$ -unbiased under any design if $b = 1$ in the (b, g) -model, but only under the design pps if $b \neq 1$. As an illustration, for gamma populations with $f = 1$ and 5, and $p_i(x) \propto x^m$ ($i = 1, \dots, n$), Table 2 gives the relative $p\xi$ -bias of T_{MR} , $E_{p\xi}(T_{MR})/\mu_Y - 1$, under deviations from $b = 1$ and pps ($m=1$). The larger the skewness, the higher is generally the bias. The pattern would be similar for T_R . Hansen, Madow and Tepping (1978) and others have pointed out the weakness in the use of model-dependent methods when the assumed model breaks down. Their conclusion is reinforced by Table 2. Another discussion pertinent to bias resulting from model breakdown is given in J.N.K. Rao (1978).

TABLE 2. $p\xi$ -bias of T_{MR} , divided by μ_Y , as a function of the model parameter b , the design function parameter m , and the population shape parameter f

b	f=1				f=5			
	m=0	m=1	m=2	m=3	m=0	m=1	m=2	m=3
0	∞	0	-0,50	-0,67	0,25	0	-0,17	-0,29
0,5	1,00	0	-0,25	-0,38	0,11	0	-0,08	-0,15
0,9	0,11	0	-0,05	-0,08	0,02	0	-0,02	-0,03
1,0	0	0	0	0	0	0	0	0
1,1	-0,09	0	0,05	0,09	-0,02	0	0,02	0,03
1,5	-0,33	0	0,25	0,46	-0,09	0	0,08	0,16
2,0	-0,50	0	0,50	1,00	-0,17	0	0,17	0,33

When $b \neq 1$, and m increases away from one, the bias of (x^m, T_R) and (x^m, T_{MR}) will increase in absolute value. Hence, the estimators T_R and T_{MR} are caught in a dilemma: if the model with $b = 1$ is true, one should, by Remark 1, choose a design with a large m in order to decrease the variance. But if the model is wrong and $b \neq 1$, the same large m brings about a penalty in the form of $p\xi$ -bias. Moreover, this bias outweighs the usually modest advantage derived from decreased variance.

The conservative, robust approach is therefore the one advocated in classical survey sampling texts, that is, to use p -unbiased or approximately p -unbiased procedures. This means the srs design for T_R , the pps design for T_{MR} . This general outlook on survey sampling, where robustness is guaranteed by the randomized selection prescribed by an (approximately) p -unbiased procedure is in general agreement, for example, with the view in Godambe & Thompson (1977).

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